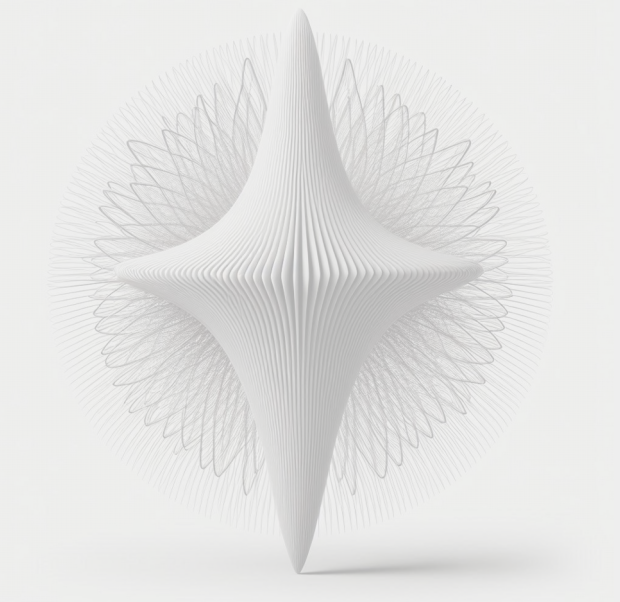


Signals and Systems

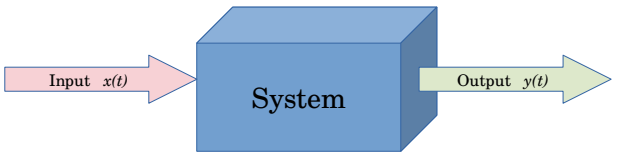
Lakshmi Narasimhan
IIT Madras

Disclaimer: There could be errors and typos in these slides. Use with caution.

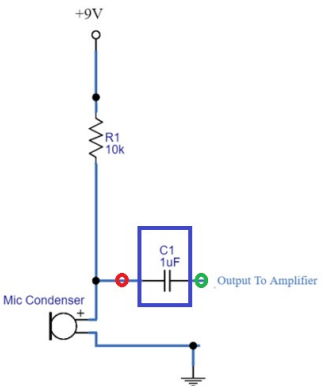
Systems

Systems (Flashback)

- A *box* that processes an input (single) signal and produces an output (single) signal



- Example
  - Input: a voltage signal
  - Output: signal without DC component



- Analytically: $f\{x(t)\} = y(t)$

Standard Systems

- Integrator: $y(t) = f\{x(t)\} = \int_{-\infty}^{t} x(v)dv$
  - $y[n] = \Sigma_{k = -\infty \text{ to } n}\, x[k]$
- Differentiator: $y(t) = f\{x(t)\} = dx(t) / dt$
  - $y[n] = x[n] - x[n - 1]$
- Time shifter: $y(t) = f\{x(t)\} = x(t - T)$
- Amplifier: $y(t) = f\{x(t)\} = Ax(t)$

Properties of Systems

Causality

- Causal:
  - Output at time t depends only on the input at time t or earlier
    - Present output depends only on past and present inputs
- Non-causal:
  - Output at time t depends on future inputs as well
- All practical systems are causal
  - Some non-causal systems can be implemented with delay

Causality - Examples

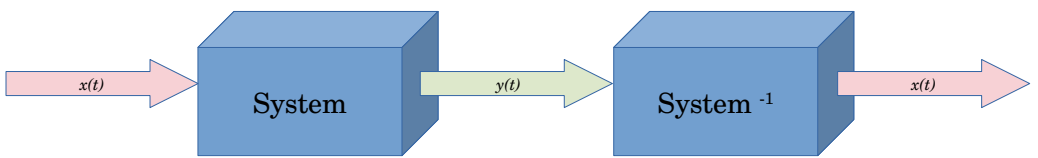
- Determine causality of the following systems
  - $y(t) = 0.3x(t)$ causal
  - $y(t) = (x(t) + x(t-1))/2$ causal
  - $y(t) = (x(t) + x(t+1))/2$ non causal
  - $y[n] = x[-n]$ non causal
  - $y(t) = x(t+T)$ non causal if $T>0$, else causal
  - $y(t+T) = x(t)$ causal if $T>0$, else non causal

Memory of Systems

- Memory less:
  - Output at time t depends only on the input at time t
  - Example: amplifier, $y(t) = (t+1)^2 x(t)$
- System with memory:
  - Output at time t depends on the past (or future) input
  - Example: integrator/differentiator (CT and DT)
    - $y(t) = x(t-1)$, $y(t) = x(at)$ for $a \neq 1$, $y[n] = y[n-1] + x[n]$

Invertibility

- Invertible system
  - Given an output signal, the corresponding input can be identified
  - For $f\{x(t)\} = y(t)$, $f^{-1}\{\ \}$ exists such that $f^{-1}\{y(t)\} = x(t)$
    - The inverse operation is another (invertible) system
  - Invertible systems are lossless transformations



Invertibility - Examples

- Check which of the following are invertible and non-invertible systems
  - $y(t) = x(t) + c$ invertible
  - $y(t) = A\,x(at + b)$ inoertible iff $A, a \neq 0$
  - $y(t) = t^2\,x(t) + c$ inoertible except at $t = 0$
  - $y(t) = \cos(x(t) + c)$ non invertible.
  - $y(t) = dx(t)/dt$ non invertible
  - $y(t) = x(t)\,x(t + T)$ non insertible

Stability

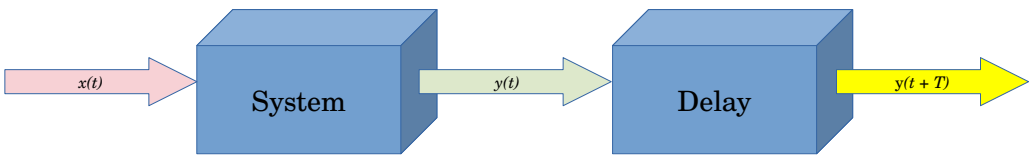
- Stable systems
  - BIBO – bounded input produces bounded output
  - For $|x(t)| < \infty$, a stable system $f\{\}$ gives an output such that $|f\{x(t)\}| < \infty$
  - For $|x[n]| < \infty$, a stable system $f\{\}$ gives an output such that $|f\{x[n]\}| < \infty$
- Critical to ensure practical systems are stable
  - An invertible system is not necessarily stable and vice versa
    - Examples: $y(t) = (t^2 + 1)\, x(t)$, $y(t) = x^2(t)$

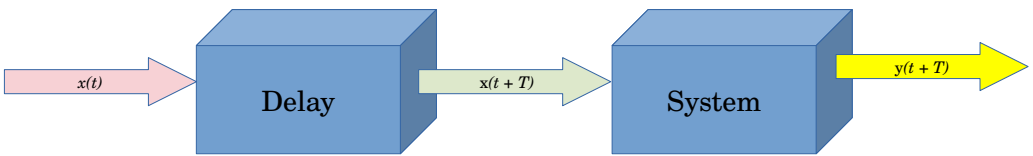
Stability - Examples

- Check which of the following are stable and unstable systems
  - $y(t) = x(t) + c$ Stable
  - $y(t) = A\,x(at + b)$ Stable
  - $y(t) = t^2\,x(t) + c$ unstable
  - $y(t) = \tan(x(t) + c)$ unstable
  - $y(t) = dx(t)/dt$ unstable
  - $y(t) = x(t)\,x(t + T)$ Stable

Time Invariance

- Time invariance
  - The system description (input-output mapping) does not change with time
    - A delayed or advanced input should produce an output which is also delayed or advanced, respectively
      - For time invariant $f\{\ \}$, if $f\{x(t)\} = y(t)$, then $f\{x(t + T)\} = y(t + T)$
  - Equivalent setups:





Time Invariance

- Time varying systems
  - Systems have different input-output mapping at different times
  - For time varying $f\{\ \}$, $f\{t, x(t)\} = y(t)$ and $f\{t + T, x(t + T)\} \neq f\{t, x(t)\}$
- Examples:
  - $y(t) = e^{-t} x(t)$
  - $y(t) = \int_{-\infty}^{t} x(v)\, dv$

Time Invariance - Examples

- Check which of the following are time invariant and time varying systems
  - $y(t) = x(t)\,u(t)$ → time variant
  - $y(t) = A\,x(at + b)$ → time variant
  - $y(t) = t^2\,x(t) + c$ → time variant
  - $y(t) = \tan(x(t) + c)$ time invariant
  - $y(t) = dx(t)/dt$ time invariant
  - $y(t) = x(t)\,x(t + T)$ time invariant

Time Invariance - Examples

- Check which of the following are time invariant and time varying systems
  - $y[n] = x[n] \cos(wn)$ variant (invariant if $w \neq 0, 2k\pi$)
  - $y[n] = A\, x[n^2 + N]$ $x[n+N_0] = Ax[n^2+N+N_0]$, $y[n+N_0] = Ax[(n+N_0)^2+N]$ $\Rightarrow$ variant
  - $y[n] = n\, x[n]$ variant
  - $y[n] = A\, x[n] + c$ invariant

Linearity

- Linear systems
  - A system that satisfied superposition
    - Sum of scaled inputs produce sum of scaled corresponding outputs
      - $f\{ax_1(t) + bx_2(t)\} = a\, f\{x_1(t)\} + b\, f\{x_2(t)\}$

Linearity - Examples

- Check which of the following (and their DT equivalents) are linear systems

linear time variant

- $y(t) = x(t)\, u(t)$

$y_1(t) = x_1(t)\, u(t) \qquad y_2(t) = x_2(t)\, u(t)$

linear time variant

- $y(t) = A\, x(at + b)$

linear time invariant

- $y(t) = dx(t)/dt$

Linearity - Examples

- Check which of the following are linear systems
  - $y(t) = t^2\, x(t) + c$ linear if $c = 0$
  - $y(t) = \tan(x(t) + c)$ not linear
  - $y(t) = x(t)\, x(t + T)$ not linear
  - $y(t) = \mathrm{Re}\{x(t)\}$ linear

Properties of RLC Systems

			
Causal	Yes	Yes	Yes
Memory	No	Yes	Yes
Invertible	Yes	No	No
Time invariant	Yes	Yes	Yes
Linear	Yes	Yes	Yes

LTI Systems

Representing a Signal with Dirac Deltas

- Recall that $\int \delta(t - v)\, x(t)\, dt = x(v)$
  - Equivalent to representing $x(t)$ as an infinite sum of scaled Dirac deltas

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(\tau - t)d\tau$$

- What if a system is linear and we know its output to be $\delta(t - v)$ for all v?

Linearity is Good

- If $h_v(t)$ is the output of a linear system for input $\delta(t - v)$, then for input $x(t)$, we get

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(\tau - t)d\tau$$

$$f\{\delta(t-\tau)\} = h_\tau(t)$$

$$f\{x(\tau)\delta(t-\tau)\} = x(\tau)h_\tau(t)$$

$$f\left\{\int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau\right\} = f\{x(t)\} = y(t) = \int_{-\infty}^{\infty} x(\tau)h_\tau(t)d\tau$$

- Linearity simplifies analysis; but, can this be made even simpler?

Linearity & Time Invariance is Better

- If $h_v(t)$ is the output of a linear & time invariant system for input $\delta(t - v)$, then $h_v(t) = h_0(t - v)$ and

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h_\tau(t) d\tau = \int_{-\infty}^{\infty} x(\tau) h_0(t - \tau) d\tau$$

- Thus, only the output of the system for $\delta(t)$ needs to be known
  - Impulse response of an LTI system $f\{\delta(t)\} = h(t)$
  - The output of any another signal $x(t)$ can be computed as $\int x(\tau)\, h(t - \tau)\, d\tau = y(t)$

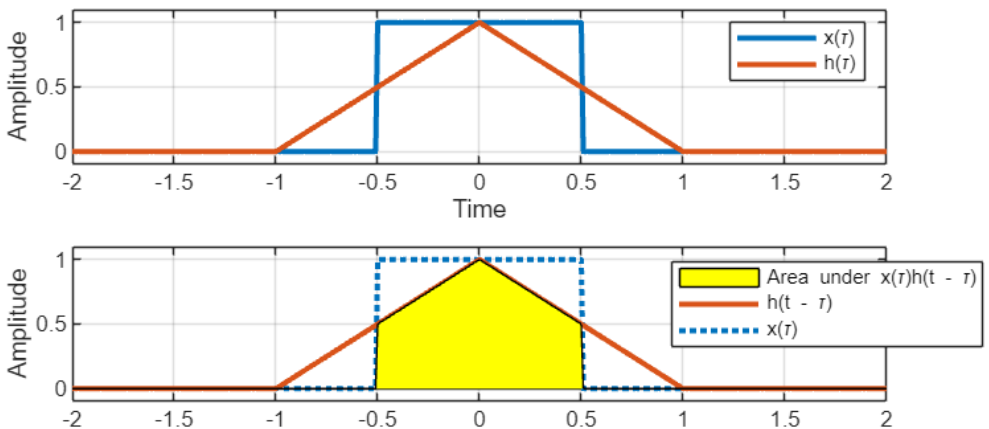
The Convolution Integral

- If $h(t)$ is the impulse response of a system, then the output of only LTI systems for input $x(t)$ can be given by $y(t) = x(t) * h(t)$
  - * represents the convolution operation $\int x(\tau)\, h(t - \tau)\, d\tau$
- Given $x(t)$ and $h(t)$, how to convolve them?
  - Step 1: Time reversal: $h(-\tau)$
  - Step 2: Time shift: for each value of t, $h(t - \tau)$
  - Step 3: Multiply and add: $x(\tau)\, h(t - \tau)$ and $\int x(\tau)\, h(t - \tau)\, d\tau$

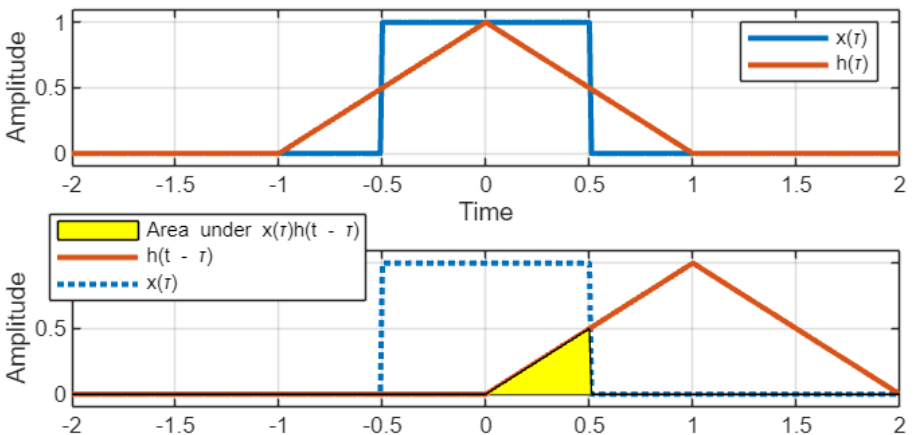
The Convolution Integral - Example

- Let $x(t) = \text{rect}(t)$ and $h(t) = \text{tri}(t)$

At t = 0, y(0) = 0.75



At t = 1, y(1) = 0.125



The Convolution Integral - Demo

- Convolution video for rect(t) * tri(t)

The Convolution Integral - Demo

- Convolution video for rect(t) * x(t)

Convolution - Examples

- For an LTI system with impulse response rec(t), find the output for input rect(t)

Convolution - Examples

- For an LTI system with impulse response $e^{-t}\,u(t)$, find the output for input u(t)

Convolution - Examples

- For an LTI system with impulse response rec(t), find the output for input tri(t)

Convolution - Examples

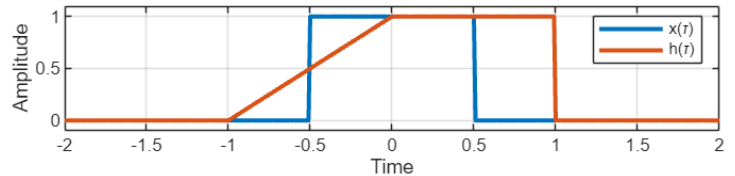
- For an LTI system with impulse response rec(t), find the output for input tri(t)

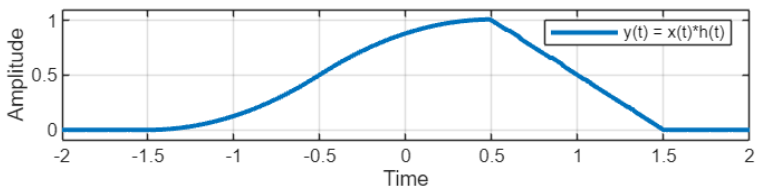
$$y(t) = \begin{cases} \int_{-0.5}^{t+1}(1+t-\tau)d\tau & -1.5 \le t \le -0.5 \\ \int_{t}^{0.5}(1+t-\tau)d\tau + \int_{-0.5}^{t}(1-t+\tau)d\tau & |t| < 0.5 \\ \int_{t-1}^{0.5}(1-t+\tau)d\tau & 0.5 \le t \le 1.5 \\ 0 & \text{elsewhere} \end{cases}$$

$$y(t) = \begin{cases} \frac{(1+t+0.5)^2}{2} & -1.5 \le t \le -0.5 \\ 3/4 - t^2 & |t| < 0.5 \\ \frac{(1-t+0.5)^2}{2} & 0.5 \le t \le 1.5 \\ 0 & \text{elsewhere} \end{cases}$$

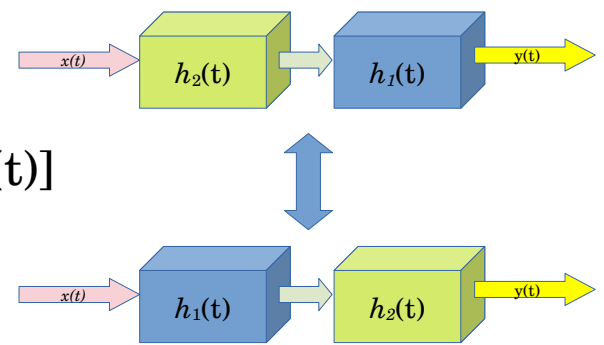
Properties of Convolution

- Time length of convolved signal
  - If $x(t)$ is T_1 long and $h(t)$ is T_2 long, then
  - $x(t)*h(t)$ is a maximum of $T_1 + T_2$ long





- Convolution is commutative and associative
  - $x(t)*h(t) = h(t)*x(t)$ and $[x(t)*h_1(t)]*h_2(t) = x(t)*[h_1(t)*h_2(t)]$
    - Order of LTI subsystems do not matter



Properties of Convolution

- Convolution is distributive: $x(t)^* [h_1(t) + h_2(t)] = x(t)^*h_1(t) + x(t)^*h_2(t)$
  - Follows from linearity
- Convolution with impulse (Dirac delta): $\delta(t)^*x(t) = \int \delta(t - v)\, x(v)\, dv = x(t)$
  - $\delta(t - T)^*x(t) = x(t - T)$
  - Shifting property
    - If $x(t)^*h(t) = y(t)$, then $x(t)^*h(t - T) = x(t)^*h(t)^*\delta(t - T) = y(t - T)$

Properties of LTI Systems

- ONLY for LTI Systems, output = input * impulse response
- Memory
  - An LTI system is memoryless if and only if its impulse response is $A\delta(t)$
- Causality
  - An LTI system is causal if and only if its impulse response is zero for $t < 0$

Properties of LTI Systems

- Invertibility
  - An LTI system is invertible if there exists $h^{-1}(t)$ such that $h(t)*h^{-1}(t) = \delta(t)$

- Stability
  - An LTI system is stable if its impulse response is absolutely integrable
  - $\int |h(t)|\ dt < \infty$
  - If $h(t)$ is periodic, then the system is unstable

LTI Systems - Examples

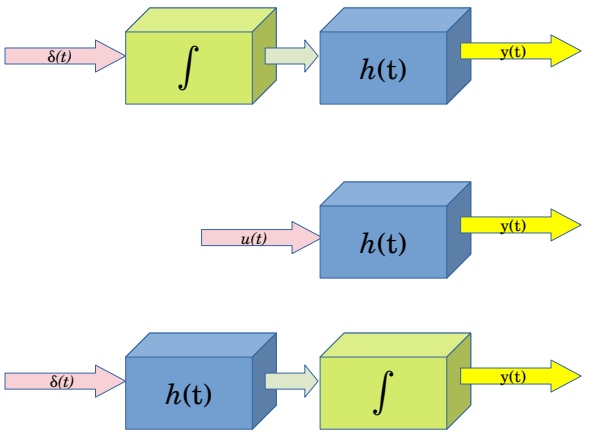
Impulse Response $h(t)$	Memory	Causal	Invertible	Stable
$e^{2t}u(1-t)$				
$e^{-\lvert t\rvert}$				
$\cos(wt)$				
$7\delta(t)$				
$(1/2)^t u(t)$				
$\Sigma_{n=-\infty}\ \delta(t - nT)$				
$\sin(t)/t$				

LTI Systems - Examples

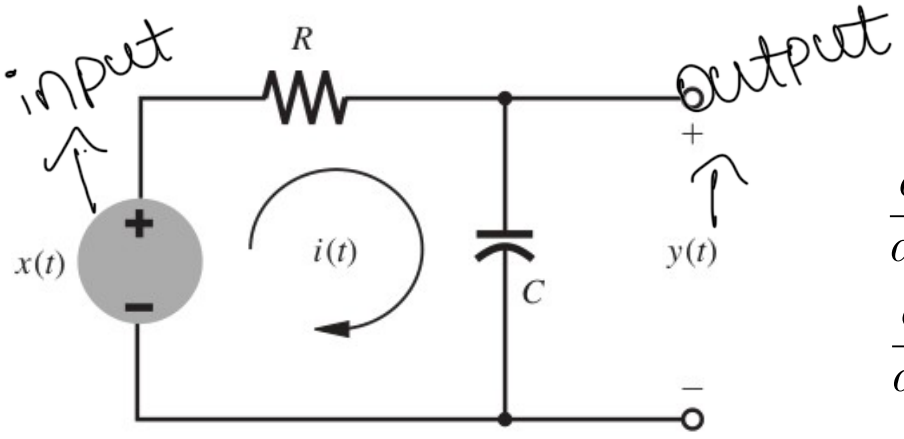
Impulse Response $h(t)$	Memoryless	Causal	Invertible	Stable
$e^{2t}u(1-t)$	No	No	Yes	Yes
$e^{-\|t\|}$	No	No	Yes	Yes
$\cos(wt)$	No	No	No	No
$7\delta(t)$	Yes	Yes	Yes	Yes
$(1/2)^t u(t)$	No	Yes	Yes	Yes
$\Sigma_{n=-\infty} \delta(t - nT)$	No	No	No	No
$\sin(t)/t$	No	No	No	No

Unit Step Response of LTI Systems

- Output of an LTI system for unit step input signal
  - Impulse response is derivative of unit step response
  - Causality: unit step response is zero for t < 0
  - Stability: unit step response is absolutely integrable
    - Sufficient but not necessary



LTI Systems - Example



$$x(t) = \frac{1}{C}\int_{-\infty}^{t} i(v)dv + Ri(t) = y(t) + Ri(t)$$

$$\frac{d}{dt}x(t) = \left(\frac{1}{C} + R\frac{d}{dt}\right) i(t)$$

$$\frac{d}{dt}y(t) = \frac{1}{C}i(t)$$

$$\boxed{x(t) = \left(1 + RC\frac{d}{dt}\right) y(t)}$$

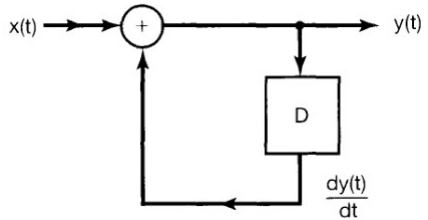
$x(t) = v_R(t) + v_C(t)$

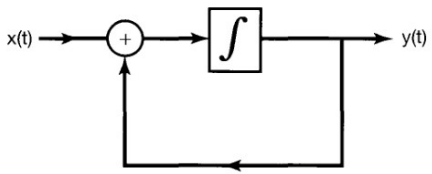
$= Ri(t) + y(t) \Rightarrow \frac{dx}{dt} = \left(\frac{1}{C} + R\frac{d}{dt}\right) i(t)$

$y(t) = \frac{1}{C}\int_{-\infty}^{t} i(\theta)d\theta \Rightarrow \frac{dy}{dt} = \frac{1}{C} i(t)$

substitute $i(t)$ here

LTI Systems - Example





Grand Summary of Systems

Summary

- Continuous-time single-input-single-output systems
  - Memory: Dependence of present output on past input
  - Causality: Present output depends only on past and present inputs
    - All practical systems are causal
  - Invertibility: One-to-one map between input and output signals
  - Stability: Bounded input produces bounded output (BIBO)
  - Examples: Integrator, differentiator, amplifier, delay (time shifter)

Summary

- Linear and time invariant (LTI) systems
  - Linearity: Satisfies superposition property
  - Time invariance: A delay input produces a delayed output
    - All passive electric circuits are LTI
  - Impulse response: Output of a system to Dirac delta input
    - For LTI: Causal if impulse response is zero for $t < 0$
    - For LTI: Stable if impulse response is absolutely integrable

Summary

- Convolution: Weighted sum of product of time reversed + shifted signal
  - Input-out relationship of an LTI system only: $y(t) = x(t) * h(t)$
  - Impulse response of the cascade of an LTI system and its inverse = Dirac delta
  - Commutative, associative, and distributive
    - Unit step response is the integral of impulse response
      - Defined when the response is bounded
- Complex exponentials are eigenfunctions of convolution (& LTI systems)